

**In this question you must show all stages of your working.**  
**Solutions relying entirely on calculator technology are not acceptable.**

The depth of water,  $D$  metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where  $t$  is the number of hours after **midnight**.

- (a) Show that the depth of water in the harbour at 2 am is just over 4 metres. (1)
- (b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres. (5)